Model variants: logistic regression, linear SVM, RBF SVM, XGBoost

FINAL_V1 experiment group summary

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Model variants over all completed representations

- Logistic regression: 14 conditions, median AUROC 0.996, median AUPRC 0.997, median speaker errors 1.
- Linear SVM: 14 conditions, median AUROC 0.995, median AUPRC 0.996, median speaker errors 1.
- RBF SVM: 14 conditions, median AUROC 0.975, median AUPRC 0.986, median speaker errors 3.
- XGBoost: 14 conditions, median AUROC 0.997, median AUPRC 0.998, median speaker errors 1.

Paired comparison with logistic regression

- Linear SVM minus logistic regression (same representation): 14 comparisons, median AUROC difference -0.001 (range -0.015 to 0.005); interval excludes zero in 0 (positive 0, negative 0). Unadjusted, descriptive.
- RBF SVM minus logistic regression (same representation): 14 comparisons, median AUROC difference -0.023 (range -0.131 to 0.007); interval excludes zero in 2 (positive 0, negative 2). Unadjusted, descriptive.
- XGBoost minus logistic regression (same representation): 14 comparisons, median AUROC difference -0.001 (range -0.007 to 0.007); interval excludes zero in 0 (positive 0, negative 0). Unadjusted, descriptive.

Reading the table

- Primary metrics are pooled speaker-level out-of-fold AUPRC and AUROC (47 speakers: 25 PD, 22 controls; prevalence 53.2% PD); F1, sensitivity, specificity and balanced accuracy use the fixed threshold 0.5.
- Intervals are percentile bootstrap intervals over speakers (2,000 resamples); with 47 speakers one person moves a metric by roughly 2 points. Comparisons are descriptive; no significance is claimed.
- Recording date and class are confounded in this corpus (every date holds one class; all 44.1 kHz audio is PD), so a high score does not show disease detection.
- The SVM pseudo-probability is the logistic function of the decision value, not a calibrated probability; the 0.5 threshold equals the SVM margin.

Table 1: AUPRC / AUROC (pooled, speaker level) by representation and model

Representation	Logistic regression	Linear SVM	RBF SVM	XGBoost
AC	1.000 / 1.000	0.998 / 0.998	0.991 / 0.989	1.000 / 1.000
AC/FORWARD	0.997 / 0.996	0.997 / 0.996	0.997 / 0.996	0.997 / 0.996
AC/BACKWARD	0.990 / 0.987	0.994 / 0.993	0.990 / 0.985	0.996 / 0.995
WLM	0.996 / 0.995	0.996 / 0.995	0.887 / 0.873	0.998 / 0.998
W2V	0.992 / 0.989	0.986 / 0.975	0.997 / 0.996	0.989 / 0.982
XLSR	1.000 / 1.000	0.997 / 0.996	0.986 / 0.976	0.997 / 0.996
H1	0.998 / 0.998	0.991 / 0.987	0.901 / 0.867	0.998 / 0.998
H2	0.997 / 0.996	0.989 / 0.982	0.956 / 0.962	0.998 / 0.998
H3	1.000 / 1.000	1.000 / 1.000	0.985 / 0.973	0.998 / 0.998
H4	0.996 / 0.995	0.992 / 0.989	0.986 / 0.975	0.994 / 0.993
H5	0.997 / 0.996	0.998 / 0.998	0.986 / 0.975	0.993 / 0.991
H6	1.000 / 1.000	0.992 / 0.989	0.986 / 0.975	0.998 / 0.998
H7	0.997 / 0.996	0.997 / 0.996	0.984 / 0.971	0.993 / 0.991
H8	0.997 / 0.996	0.998 / 0.998	0.986 / 0.975	0.998 / 0.998